Manufacturing

The manufacturing phase transitions the project from detailed design to physical reality. Building upon the comprehensive documentation developed in the Detailed Design Phase, this phase focuses on executing controlled manufacturing and procurement processes to deliver components that meet all specified requirements. The objective is to ensure every component functions as expected when tested in isolation, providing confidence before integration into the final assembly.

By the end of this phase, each project should have a manufactured or procured component that has been verified through testing and is ready for assembly. It is expected that components will behave as designed when fully integrated with the broader system. The next phase, Assembly & Testing, will validate this integration and ensure all components work cohesively.

To get manufacturing approved, the Manufacturing Review Process is outlined here <link>

	Deliverable	**Criteria**
1	**Project**	
1.1	Project Plan Document	Create a comprehensive project plan detailing the manufacturing process, including scope, milestones, part list, timeline, dependencies, constraints, requirements, budget, risk analysis, change management, and bill of materials. Transfer relevant documentation from the Detailed Design Phase.
1.2	Chief Engineer Approval	Obtain approval from the Chief Engineer before commencing manufacturing or procurement activities. This applies to all major assemblies.
2	**Manufacturing & Construction**	
2.1	Quality Control Plan	Develop a Quality Control Plan for each component, documenting project information, timeline, requirements, quality control methods, inspection procedures, measurement tools, acceptance criteria, and sampling plans. This establishes standards for ensuring the component meets specifications.
2.2	Chief Designer Approval	Obtain approval from the Chief Designer for the Quality Control Plan before proceeding with manufacturing.
2.3	Manufacturing Execution	Execute the physical creation and assembly of components, adhering strictly to engineering drawings and specifications. Document any deviations immediately, maintain material and process traceability, follow discipline-specific

		manufacturing procedures and safety protocols, and coordinate with other teams on interfacing components. Refer to your Discipline Guide for detailed manufacturing processes.
3	**Procurement**	
3.1	SLT Approval	Submit procurement request with complete documentation and reasoning to the department lead, then obtain Senior Leadership Team (SLT) approval. The SLT will verify the request meets project requirements, is cost and time effective, and check for existing tools, parts, or materials.
3.2	Purchase & Delivery	Following approval, the procurement officer places the order with the selected supplier, coordinates delivery logistics and timeline, and monitors order status.
3.3	Inventory Management	Upon delivery, the procurement officer receives the item, transfers it to the workshop, and notifies the department member. Verify the item matches the order (quantity, specifications, condition), log it into the inventory system with SKU, and maintain accurate purchase records.
4	**Testing & Verification**	
4.1	Test Report	Create a Test Report for each component to verify it meets requirements and behaves as expected in isolation. For ADR-related testing, follow tailored test documents provided by Systems Engineers.
4.1.1	Scope	Document component identification, intended purpose, project leads, test conductors, and test environment and conditions.
4.1.2	Data Collection Plan for Acceptance Testing	Outline the data collection process, including type of data (numerical, categorical, qualitative), collection methods and procedures, collection frequency, responsible personnel, rationale for each test, description of ideal/expected results, and pass/fail criteria.
4.1.3	Data Analysis Report	Provide a summary of data and insights from analysis, including analytical approach used, trends and patterns identified, statistical analysis where applicable, conclusions drawn

		from results, recommendations based on findings, and comparison to requirements and expected performance.
4.1.4	Updated Component Specifications	Document updated specifications reflecting final requirements based on test results. Note any changes to performance parameters, material properties, or requirements that need adjustment based on actual performance.
4.1.5	Appendix	Include visual documentation (equipment setup photographs, test in progress documentation, results visualization, failure modes if applicable) and raw data (unprocessed data in original format, test logs and measurement records, sensor readings or output files, supporting calculations).
4.2	Systems Liaison Approval	Obtain approval from the project's Systems Liaison for all Test Reports. Each test requires a separate report serving as evidence that the component meets requirements.